

Can Fundamentals Alone Predict Market Reactions? A Machine Learning Approach to Post-Earnings Stock Movement in Indian Sectors

Interim Report

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Introduction

In Indian financial markets, Quarterly earnings announcements play a critical role in short-term stock price behavior. These disclosures often trigger significant short-term stock price movements; a dynamic that academic research has captured through phenomena like the post-earnings announcement drift (PEAD). However, while it's widely accepted that financial fundamentals such as earnings per share (EPS), profit after tax (PAT), and operating profit margin (OPM%) are central to these movements, most predictive models still rely on a mix of technical indicators, sentiment data, or textual analysis. This makes it difficult to isolate the true informational value of structured financial disclosures.

This study addresses that gap. This study investigates whether structured, numerical financial indicators from quarterly earnings reports, closing price of T, T+1, T+2 and T+3 without any additional data sources can predict the direction of stock price movement (up or down) in the three trading days following the announcement. To explore this, we apply supervised machine learning models (Logistic Regression, Random Forest, and XGBoost) to data from 45 listed Indian companies across five sectors.

What sets this work apart is its strict focus on structured data, 3 day closing price and its cross-sector analysis for Indian market only. Unlike prior research: Huanget al.(2021), Sherasiya and Jadeja (2023) that either targets long-term trends or uses blended inputs, Zhao et al. (2023) provided evidence that financial features could yield robust models in emerging markets like China but also relied on price-based features. Ampomah et al. (2020) compared ensemble methods but did not isolate the post-earnings effect, while Saberironaghi et al. (2025) provided a comprehensive survey highlighting the under-utilization of earnings-based triggers in short-term forecasting. We will also compare

model performance and assess which financial indicators most influence predictions using interpretation tools like SHAP or permutation importance.

In doing so, the study contributes a transparent, replicable framework for short-term stock movement prediction and clarifies how much insight quarterly fundamentals truly provide, sector by sector, model by model.

Scope and objectives

Scope

This study focuses on listed Indian companies across five selected sectors, aiming to predict short-term stock price movement (Up/Down) in the three trading days post quarterly earnings announcements. The analysis exclusively uses publicly available, structured numerical financial data(e.g., EPS, PAT, OPM%) and does not include unstructured inputs such as sentiment, analyst commentary, news, or AI-generated content. This study intentionally excludes unstructured inputs to isolate the predictive power of structured financial disclosures.

Short-term stock movement refers to the price change direction (Up/Down) over three trading days following the earnings announcements(T+1 to T+3). Five sectors will be chosen to ensure diversity in market dynamics and will represent a unique combination of market capitalization, and availability of quarterly disclosures.

Objectives

The central objective of this study is to evaluate whether structured financial disclosures such as earnings per share (EPS), profit after tax (PAT), and operating profit

margin (OPM%) can, on their own, predict short-term post-earnings stock price direction in the Indian equity market. By focusing exclusively on structured numerical data, the study deliberately sets aside unstructured inputs like sentiment or analyst commentary, to isolate the “true” informational value of fundamentals. In doing so, it not only measures predictive capability but also compares model performance across sectors and algorithms, while identifying the financial indicators that most strongly shape outcomes.

To achieve this, the study is guided by four interlinked research questions, each addressed through a systematic methodology. The first question (RQ1) asks whether structured quarterly indicators alone can predict stock price movements in the three trading days following earnings announcements. This will be tested using supervised classification models—Logistic Regression, Random Forest, and XGBoost—trained on labelled financial data. Performance will be evaluated through widely accepted metrics such as accuracy, precision, recall, and F1-score, offering a clear assessment of predictive validity.

The second question (RQ2) digs deeper by examining *which* financial indicators matter most. Are investors more responsive to profitability measures such as EPS, or to efficiency ratios like OPM%? To answer this, model interpretation techniques such as SHAP (Shapley Additive Explanations) and permutation importance will be applied. These tools go beyond prediction to reveal transparency, showing which features consistently drive model decisions and therefore carry the most weight in influencing market reactions.

Since market dynamics differ across industries, the third question (RQ3) investigates whether predictive performance is uniform or sector dependent. By stratifying the test dataset across five sectors—IT, BFSI, FMCG, Pharma, and Auto—the study will evaluate whether structured indicators have stronger signals in some industries than in

others. Sector-level classification results, measured primarily with accuracy and F1-score, will help clarify whether fundamentals play a universal role or one shaped by sector-specific investor behaviour.

Finally, the fourth question (RQ4) compares the relative strengths of the three machine learning models themselves. Logistic Regression provides interpretability, Random Forest offers robustness to collinearity, while XGBoost is known for strong performance with structured data. By conducting stratified k-fold cross-validation and evaluating performance through accuracy, F1-score, and ROC-AUC, the study will recommend the most effective algorithm for short-term prediction.

Together, these objectives and research questions create a coherent roadmap. They not only test whether fundamentals “matter” for short horizon forecasting but also reveal *how much, where, and through which algorithms*. This layered design transforms the study into a comprehensive exploration of fundamentals-based prediction in Indian markets.

Literature survey

Research on machine learning (ML) for stock prediction has expanded rapidly, with growing interest in whether structured fundamentals alone can provide predictive signals. Huang et al. (2021) demonstrate the long-term potential of financial statement data by compiling 22 years of fundamentals for S&P100 firms. Their study shows that Random Forest and ensemble methods outperform baselines and even a DJIA buy-and-hold benchmark, confirming that financial ratios such as EPS and profit margins contain exploitable information for multi-year forecasting. Similarly, Tsai et al. (2023) align fundamental ratios for Taiwanese firms and apply advanced models including a Graph Attention Network (FinGAT). By ranking stocks on predicted returns, they build portfolios

that outperform the TW50 index by wide margins, highlighting that structured ratios can be powerful mid-to-long-term predictors when properly aligned and modeled.

Evidence from emerging markets reinforces these findings. Khanpuri et al. (2024) and Sherasiya and Jadeja (2023) examine Indian firms using quarterly and annual financial disclosures. Khanpuri et al. emphasize the potential of deep learning architectures (LSTM/RNN) on NIFTY50 data, while Sherasiya and Jadeja apply simpler classifiers (SVM and Decision Trees) to a narrower feature set. Both find that profitability metrics (EPS, OPM) and valuation ratios (P/E) play a central role in determining price trends. These studies, however, primarily target long-term horizons rather than short-term market reactions to earnings announcements.

Other contexts add further nuance. Namdari and Li (2021) show that fundamentals improve directional accuracy in U.S. technology firms, even marginally outperforming models based on price history alone. Wang (2025) demonstrates that nonlinear ML methods, especially XGBoost, are effective at forecasting major U.S. stock declines using accounting ratios, achieving ROC-AUC around 0.70 for severe one-year drops. Hanauer et al. (2022) use European accounting data to build ML-based “value spreads,” finding that boosted tree models significantly outperform traditional linear factors in generating excess returns. Zhao et al. (2023) extend the evidence to China, showing that Random Forest and boosting models trained on profitability and growth indicators provide superior forecasts of monthly returns. These cross-market studies consistently highlight the nonlinear relationships between fundamentals and returns and the advantage of ML models in capturing them.

Beyond single studies, Ampomah et al. (2020) compare a suite of ensemble models across multiple markets and conclude that Extra Trees offers robust performance across accuracy, precision, and AUC. Saberironaghi et al. (2025) provide a broader survey, cataloging ML and deep learning methods for stock prediction. They emphasize both the advances brought by architectures like LSTM and CNN, as well as persistent challenges: data quality, overfitting, and interpretability. Their review underscores that structured datasets—particularly those built from financial disclosures—remain underexplored relative to hybrid approaches that combine technical or sentiment indicators.

Taken together, the literature establishes three key insights. First, structured financial indicators such as profitability, leverage, and efficiency ratios do contain predictive signals across horizons and markets. Second, nonlinear ML approaches, particularly tree-based ensembles and deep learning, often outperform linear models by capturing complex relationships. Third, most studies focus on long- or medium-term predictions, often augmenting fundamentals with market sentiment or technical data. Very few isolate quarterly financial disclosures to study short-term post-earnings reactions, and almost none apply this lens systematically to the Indian equity market.

This creates the clear research gap addressed in the present study: whether structured fundamentals, without sentiment or technical overlays, can predict short-term post-earnings stock price direction in India. By focusing on quarterly disclosures across sectors, this project tests the limits of fundamental-only ML models in a context that has been largely overlooked in prior work.

Data Description

Every research project rests on the quality and integrity of its data. The foundation of the analysis is a raw file titled **MS_Capstone_Final_Raw_Dump.xlsx**, which contains quarterly fundamentals and market reactions for India's top 45 listed companies, spanning five key sectors—Information Technology (IT), Banking and Financial Services (BFSI), Fast-Moving Consumer Goods (FMCG), Pharmaceuticals, and Automobiles. The period covered stretches from July 2022 to July 2025, amounting to 13 quarterly result cycles and nearly **585 observations**.

Unlike many curated academic datasets, this one was intentionally left **raw**. No normalization, scaling, or feature engineering was applied at the point of extraction. Instead, the dataset is designed to preserve the original reporting style of each company and sector, ensuring full transparency and reproducibility. Downstream in this project, we will build transformations and derived features. This choice ensures that what we analyse today can be reconstructed tomorrow, an essential principle in scientific research.

What the Dataset Captures

At its core, the dataset links two types of information: structured financial disclosures and market price reactions. For each company and quarter, three categories of information are recorded:

1. **Event Timing** – the official Result Date, or earnings announcement date.
2. **Fundamentals** – numerical disclosures such as Sales, Net Profit, EPS, Operating Profit, OPM%, Interest, Tax%, and sector-specific fields such as “Revenue” and “Financing Margin%” for BFSI companies.

3. **Market Response** – the closing prices on the event day (T) and the three subsequent trading days (T+1, T+2, T+3).

From these inputs, derived variables are constructed. For example, the 3-Day Average Price Post Result is the average of T+1 through T+3 closes. The 3-Day Return (%) measures the percentage change relative to T, while the Movement Label translates this into a binary outcome (1 for Up (Bullish) if the return $\geq 1\%$, 0 for Down (Bearish) otherwise).

Each column has a clearly defined role:

- Identifiers: Company, Ticker, Alpha Vantage Ticker, Sector, Market Cap Category, Quarter Label.
- Fundamentals: Sales, Expenses, Operating Profit, Other Income, Interest, Depreciation, Profit Before Tax, Net Profit, EPS, OPM%, Revenue (BFSI), Financing Profit, Financing Margin%, and Tax %.
- Market Prices: Closing Price (T, T+1, T+2, T+3) and associated trading dates.
- Derived Targets: 3-Day Avg Price, 3-Day Return, Movement Label.

Data Sources and Extraction Process

The dataset draws from a carefully orchestrated pipeline of sources, each chosen to maximize accuracy while minimizing bias. The process follows three layers: **quarterly fundamentals, event dates, and daily market prices.**

1. Quarterly Fundamentals & Result Dates

The backbone of the fundamentals comes from **IndianAPI**, *Indian API*

Marketplace. (n.d.). <https://indianapi.in/indian-stock-market> , a structured data provider that mirrors official company disclosures. Two endpoints are used:

- *Indian API Marketplace*. (n.d.-b). https://indianapi.in/indian-stock-market#/historical_stats pulls financial statement items for each quarter exactly as reported.
- *Indian API Marketplace*. (n.d.-c). https://indianapi.in/indian-stock-market#/corporate_actions identifies board announcements tagged as “Financial Results,” used to confirm the official Result Date.

Importantly, industry-specific differences are preserved. For instance, banks disclose “Financing Profit” and “Net Interest Margin,” while non-BFSI companies focus on Sales and OPM%. This choice avoids the distortion of forcing industries into a single schema and respects the diversity of accounting practices.

2. Daily Prices

Price data is more delicate, because earnings dates often fall on non-trading days or near holidays. To ensure integrity, the study applies a **strict event-anchored alignment**:

- **T (event day)**: If the stock traded on the Result Date, that closing price is recorded. If markets were closed, the nearest previous trading day is used.
- **T+1, T+2, T+3**: These are the next three actual trading days (not calendar days). Weekends and exchange holidays are automatically skipped.

The **cascade of preference**:

- **Primary**: Official exchange “Bhavcopy” archives from BSE and NSE.

- **Secondary:** IndianAPI daily endpoints, but only if marked as *meta.is_weekly = false* to ensure daily granularity.
- **Tertiary:** Alpha Vantage *API Documentation* | *Alpha Vantage*. (n.d.). <https://www.alphavantage.co/documentation/#daily>, used only when the first two sources fail.

This three-stage sourcing ensures completeness while minimizing reliance on external aggregators.

3. Integrity Checks

Multiple safeguards are embedded in the extraction logic. Each row retains the actual trading dates used (*_T_date*, *_T1_date*, etc.), so that every price is traceable. No forward-filling is done, meaning that if data is missing even after all sources are checked, the fields remain blank and are flagged for backfilling later.

This conservative approach prevents artificial smoothing.

What Is Not Done with this raw layer

Equally important is what the dataset intentionally avoids:

- No **unit scaling**, currency conversions, or inflation adjustments. Figures are in raw reported form.
- No **deduplication** beyond unique company-quarter keys.
- No **synthetic imputation**. If a field is unavailable across all sources, it remains missing.

Analysis

Data Cleaning

Before exploring relationships in the dataset, it was essential to ensure that the raw financial and price information was both consistent and reliable. The data quality checks were carried out using a structured Python workflow (*Jupyter Notebook Data_Quality_Check_Summary.ipynb*) that focused on four main aspects: standardization, type conversion, integrity validation, and documentation of issues.

The first step involved **standardizing raw entries**. Text fields were trimmed of stray spaces, while placeholders such as "None", "NaN", and em-dashes were uniformly replaced with missing values. This step ensured that missingness was recognized consistently across the dataset, avoiding hidden biases in later analysis.

Next came **robust date parsing**. Quarterly result dates and post-announcement trading dates (T, T+1, T+2, T+3) appeared in multiple real-world formats: Excel serial numbers, slash-separated strings, and day-first notations common in India. A custom parser was built to handle this diversity, converting all dates into standardized datetime objects while logging any failures for audit. This not only preserved chronological order but also allowed for later checks of trading day alignment. All financial columns—Sales, EPS, OPM%, PAT, and BFSI-specific fields like Financing Margin—were then **safely coerced into numeric form**. Non-numeric characters such as commas, currency symbols, and percentage signs were stripped away. Where values were invalid, they were converted to missing rather than silently forced, ensuring that no misleading figures entered the analysis pipeline.

Finally, **quality control flags** were generated to detect anomalies: missing or duplicate Result Dates, non-monotonic trading sequences, prices without corresponding dates, and implausible

percentage values (e.g., Tax % outside 0–100). Rather than deleting these rows, all flagged records were preserved with annotations, balancing completeness with transparency.

The outcome of this process was a cleaned dataset written back to an auditable workbook *Vedwa, A. (unpublished). MS_Capstone_Final_Quality_Check_Summary [Dataset]*. This ensured that subsequent exploratory analysis was built on data that was not only statistically robust but also traceable to original disclosures.

EDA Results

For this project, EDA serves two purposes: to confirm the dataset’s statistical adequacy and reliability, and to uncover whether financial fundamentals exhibit early signals that could help predict short-term post-earnings market reactions. Each step of the analysis builds toward answering the central research question: *can fundamentals alone explain market direction?*

Target Variable: Movement Label

The dataset’s dependent variable is the **Movement Label**, a binary classification of whether the stock moved “Up” or “Down” in the three trading days following an earnings announcement. Figure 1 shows its distribution. Of 584 usable observations, **344 (59%) were Down** while **240 (41%) were Up**.

This mild imbalance suggests a slight bearish tilt in the Indian market during the sample period (2022–2025). From a modelling perspective, however, it presents a risk: without adjustments, a naïve model might predict “Down” more often and still achieve superficially high accuracy. This insight signals the need for resampling or class reweighting when training classifiers.

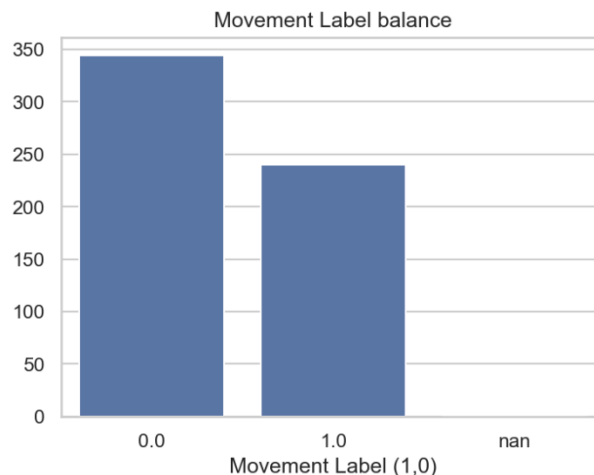


Figure 1: Movement Labels balance (Up vs. Down).

Distribution of 3-Day Returns

To understand the magnitude of post-earnings reactions, we examined the distribution of the **3-Day Return (%)**, defined as the percentage change in average closing price from T+1 to T+3 relative to the pre-announcement price (T). Figure 2 presents a histogram of this distribution.

The results reveal a market that is typically efficient but occasionally volatile. The **mean return is close to zero ($\mu \approx 0.29\%$)** with a standard deviation of **$\sigma \approx 4.28\%$** . Most reactions cluster between **-5% and +5%**, consistent with the notion that earnings announcements trigger modest adjustments rather than dramatic swings. Yet, the tails tell a different story: losses of up to -12% and gains of nearly +19% occurred, representing moments where fundamentals or expectations sharply diverged.

This duality stable centre with fat tails—highlights why machine learning is necessary. Simple linear expectations may capture average behaviour but miss the extremes that matter most to investors.

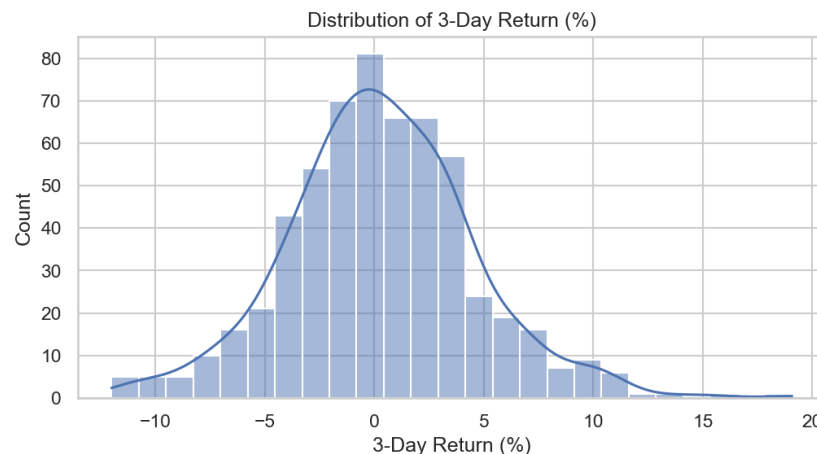
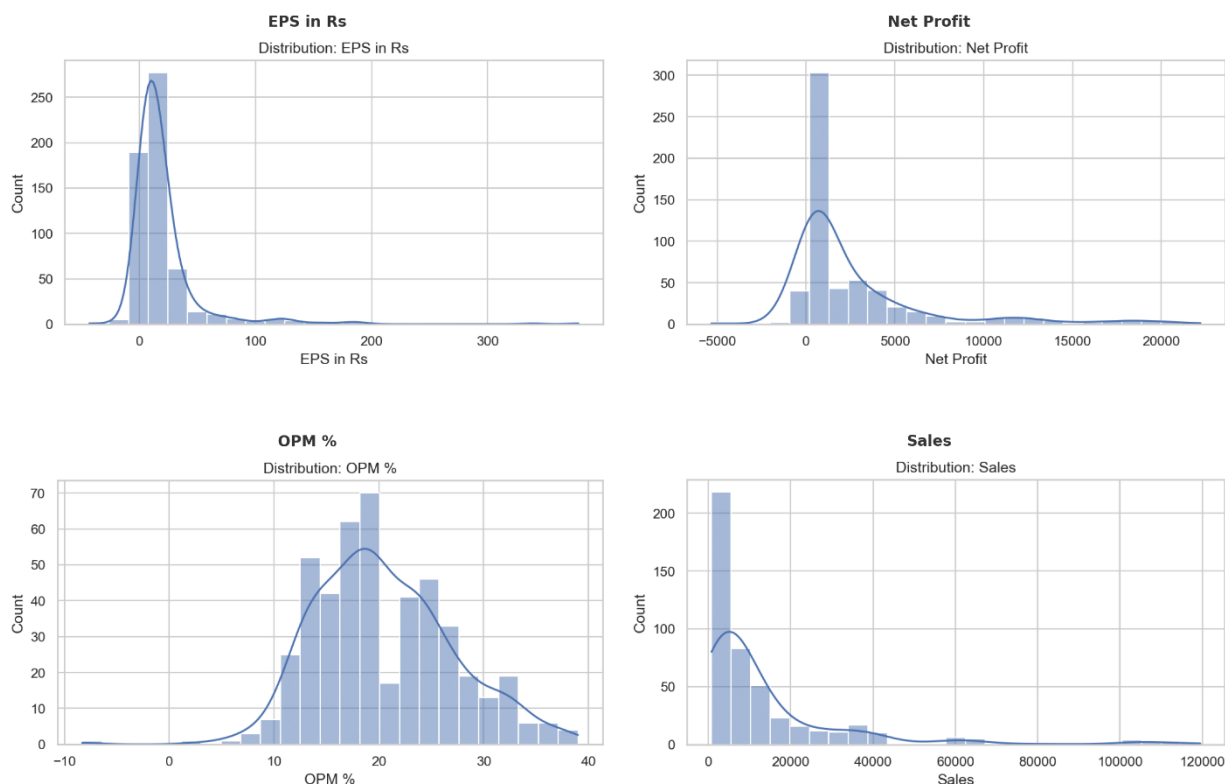


Figure 2: Histogram of 3-Day Returns (%).

Fundamentals Distributions

We next examined the underlying financial fundamentals. EPS, Net Profit, and Sales exhibit heavy-tailed distributions, while OPM% remains bounded between 0 and 100 (Figure 3). Outliers are not anomalies in the statistical sense but reflections of the diverse scale of Indian listed companies: small-cap firms with single-digit earnings coexist alongside large multinationals with thousands of crores in profit.

Interestingly, when these fundamentals were stratified by Movement Label, their distributions overlapped almost entirely. High EPS or Net Profit did not guarantee an “Up” classification. In fact, in the Auto sector, several large positive earnings surprises corresponded to “Down” movements, possibly reflecting investor disappointment relative to expectations. This early signal suggests that absolute levels of fundamentals alone may not discriminate direction, at least in raw form.



Correlation Analysis

To explore relationships systematically, we constructed a **Spearman correlation heatmap** (Figure 4). The picture that emerged was dominated by **accounting clusters**:

- Net Profit and Profit Before Tax correlated nearly perfectly ($\rho \approx 1.0$).
- Sales and Expenses mirrored one another ($\rho \approx 1.0$).
- Within BFSI, Revenue, Interest, and Financing Profit formed their own block.

These results confirm strong **multicollinearity**, which can destabilize linear models like Logistic Regression. More importantly, correlations between fundamentals and 3-Day Returns were negligible: the largest absolute value was ≈ 0.09 , close to statistical noise. This finding aligns with financial theory: fundamentals may influence markets, but the effect is subtle and often nonlinear.

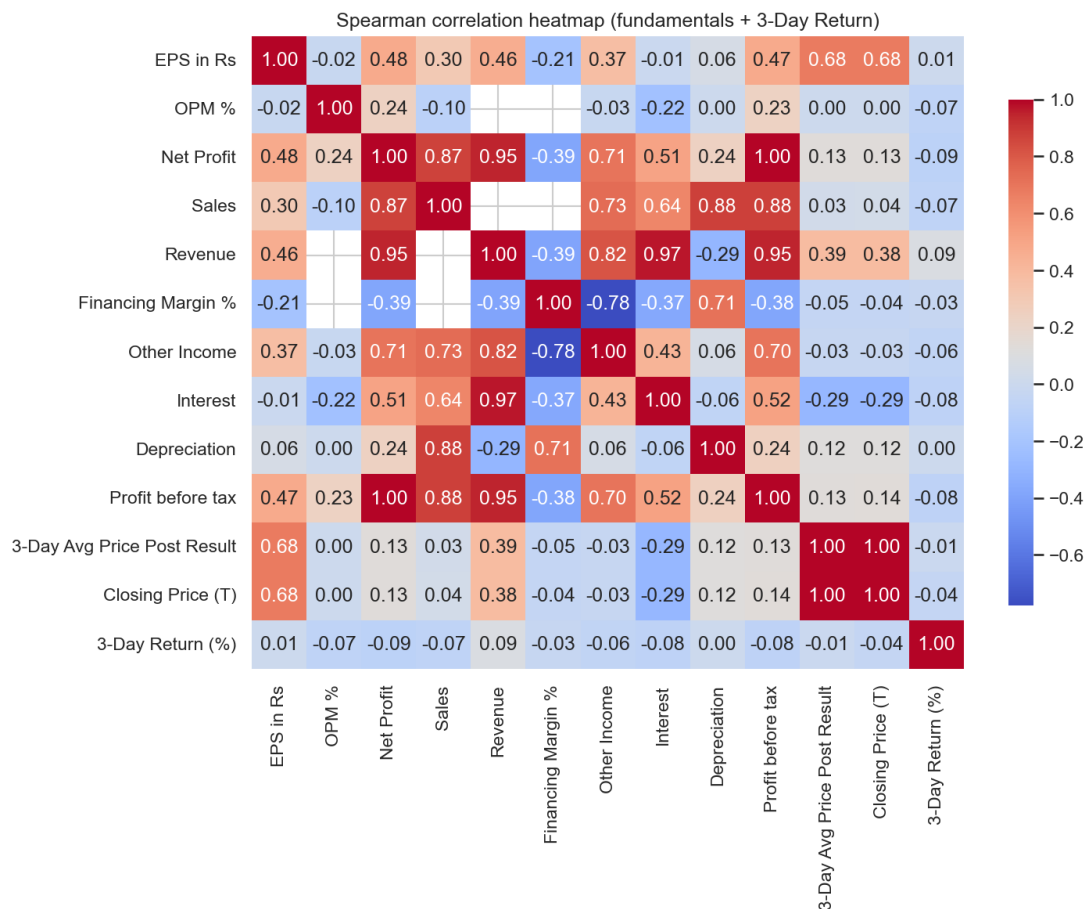


Figure 4: Spearman correlation heatmap.

Mutual Information with Movement Label

Correlation only captures linear or monotonic relationships. To probe nonlinear dependencies, we calculated Mutual Information (MI) scores between each fundamental and the Movement Label. The top scores were:

- Sales: MI ≈ 0.027
- Financing Margin %: MI ≈ 0.023
- Depreciation / Other Income: MI ≈ 0.008

These values are small, reinforcing the earlier conclusion: structured fundamentals do not provide strong standalone signals. However, subtle traces exist, particularly in BFSI-specific variables, suggesting sectoral dynamics may matter.

Top 15 Mutual Information with Movement Label

feature	mutual_info
3-Day Return (%)	0.6761504834622363
Sales	0.02692447868767789
Financing Margin %	0.022721840914435854
Depreciation	0.01809870240004363
Other Income	0.00791389031781975
Expenses	0.007482650161347637
Closing Price (T+3)	0.006939276351905388
Financing Profit	0.0061331142871818844
3-Day Avg Price Post Result	0.006059007206376421
Closing Price (T)	0.0026878699386838623
Revenue	0.0023249226075658935
Profit before tax	0.0
Interest	0.0
OPM %	0.0
Net Profit	0.0

Figure 5: Top 15 MI with Movement Label.

Sector-Level Insights

When results were split by sector, heterogeneity became more visible (Figure 6).

BFSI: average three-day returns were slightly negative (-0.45%), reflecting a cautious investor response to financial disclosures. Financing Margin %, ranging from -40% to +40%, showed no monotonic relationship with movement direction.

Auto: Exhibited overlapping distributions of EPS and Net Profit across Up and Down classes, hinting at mean-reversion effects rather than earnings-driven upside.

FMCG: Down movements often corresponded to larger top lines and slightly higher margins—suggesting scale alone does not drive performance.

IT & Pharma: medians for fundamentals were similar across Up and Down outcomes, underscoring weak separation at the level of reported figures.

The key takeaway: sector-specific dynamics shape predictability, making sector-stratified modelling not just helpful but essential.

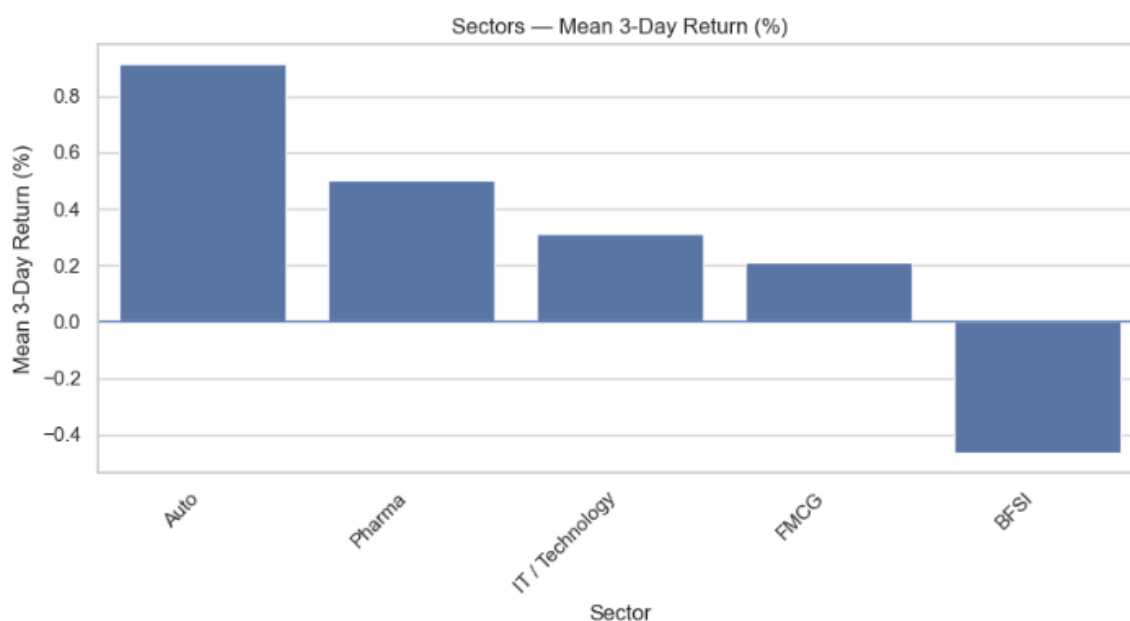


Figure 6: Average 3-Day Returns by Sector.

Outliers and Robustness

The dataset also revealed several extreme values:

- EPS and Net Profit distributions had long right tails.
- Financing Margin % swung between -39% and +39%.

- Tax % reached implausible highs (up to 3925%), likely reflecting exceptional accounting adjustments.

Feature	N (non-null)	Q1	Q3	IQR	Lower Fence	Upper Fence	Outliers (count)	Outliers (%)
Topline (Sales/Revenue)	585	3433	18790	15357	-19602.5	41825.5	59	10.09
Net Profit	585	410	3336	2926	-3979	7725	58	9.91
EPS in Rs	585	5.22	21.96	16.74	-19.89	47.07	48	8.21
3-Day Return (%)	584	2.2325	2.8075	5.04	-9.7925	10.3675	20	3.42
OPM %	468	16	24	8	4	36	9	1.92

Table 1: Outliers

Instead of removing these outliers, the study chose to retain them to preserve economic realism.

EDA shows that market reactions to earnings are modest on average but occasionally extreme, making prediction both difficult and worthwhile. Fundamentals are highly collinear and exhibit only weak direct correlations with short-term returns, limiting their standalone predictive power. Yet, subtle nonlinear signals emerge at the sector level, suggesting industry-specific dynamics influence investor responses. The dataset is mildly imbalanced (59% Down vs. 41% Up), requiring corrective measures in modelling, while outliers in EPS, Tax %, and Financing Margin reflect genuine economic events rather than errors, and are therefore retained with caution.

Overall, the analysis confirms that forecasting post-earnings stock movements using structured fundamentals alone is a formidable task, but one where machine learning is justified by subtle, sector-dependent patterns hidden in the data.

Sample Size Adequacy

A crucial step in validating the credibility of any empirical study is to confirm whether the available sample size is sufficient to support statistical inference. In this project, the inferential target is not a single population proportion, but rather the **performance of classifiers** that predict a binary outcome: whether a stock moves “Up” or “Down” in the three trading days following an earnings announcement. Accordingly, the sample size determination is grounded in **statistical power analysis for the area under the ROC curve (AUC)**, which is the study’s primary evaluation metric. Accuracy is also considered, but only as supporting intuition.

Powering on AUC Superiority over Chance

The primary test evaluates whether classifiers perform better than random chance, formalized as:

$$H_0: AUC = 0.5 \quad \text{vs.} \quad H_1: AUC > 0.5$$

with a two-sided significance level $\alpha = 0.05$ and desired power $1 - \beta = 0.80$.

Using the **Hanley–McNeil formulation** for the variance of the AUC, with n_1 positives (Up) and n_0 negatives (Down), the variance can be expressed as:

$$Var(AUC) = \frac{AUC(1 - AUC) + (n_1 - 1)(Q_1 - AUC^2) + (n_0 - 1)(Q_2 - AUC^2)}{n_1 n_0}$$

$$\text{where } Q_1 = \frac{AUC}{2 - AUC} \quad \text{and } Q_2 = \frac{2AUC^2}{1 + AUC}$$

The rejection threshold is defined by:

$$Z = \frac{AUC - 0.5}{SE} \geq z_{1-\left(\frac{\alpha}{2}\right)} + z_{1-\beta} = 1.96 + 0.842 = 2.802$$

Required and Achieved Sample Sizes

Given the observed class mix (41% Up, 59% Down), the following sample sizes are required to detect performance above chance:

- Detect AUC = 0.60 → $N \approx 265$ (≈ 109 Up, 156 Down)
- Detect AUC = 0.62 → $N \approx 182$ (≈ 75 Up, 107 Down)
- Detect AUC = 0.65 → $N \approx 113$ (≈ 46 Up, 67 Down)

The dataset at hand contains 584 usable observations ($n_1=240$ Up, $n_0 = 344$ Down). Plugging these values into the variance formula, the standard error for AUC = 0.60 is ≈ 0.024 . The resulting test statistic is:

$$Z = \frac{0.60 - 0.50}{0.02398} = 4.17$$

which comfortably exceeds the threshold of 2.802. This translates to statistical power of ≈ 0.986 , confirming that the dataset is well-powered to detect classifier performance at or above 0.60 AUC. For higher AUC values (0.62, 0.65), achieved power approaches 1.0.

Sector-Level Adequacy

Sector-level inference, however, is more constrained. With five sectors, the average sample per sector is ≈ 117 . To detect an AUC of 0.60 or 0.62, required sample sizes are ≈ 265 and ≈ 182 respectively. Thus, sector-level AUCs will be reported with confidence intervals but treated as exploratory rather than confirmatory, unless a sector's sample size exceeds 180–265.

Model Comparisons

For model-to-model performance comparisons (RQ4), bootstrap or DeLong tests for $\Delta\text{AUC} \approx 0.03\text{--}0.05$ are feasible with $N=584$. Accuracy-based comparisons (via McNemar's test) are more sample-intensive: with discordance of 20% and a gap of 6%, the required $N \approx 437$, which is achievable here. Against a majority baseline (59% Down), detecting 65% accuracy requires ≈ 518 samples—also satisfied—but detecting smaller gaps (62% vs. 59%) would require over 2,000 observations, beyond this dataset.

Conclusion

Overall, the sample size of 584 events is adequate to power the study's primary tests at the aggregate market level. The dataset can reliably detect classifier performance above chance ($\text{AUC} \geq 0.60$) and support meaningful comparisons between models. However, sector-level analyses will be interpreted descriptively with confidence intervals rather than formal hypothesis testing, given lower per-sector counts. This balance ensures the study remains both statistically rigorous and realistic in scope.

Modelling

The modelling framework is designed to operationalize the research questions while respecting the constraints and opportunities revealed during exploratory analysis. The dataset contains 584 firm-quarter observations across five sectors, with predictors limited to structured financial disclosures. The modelling strategy must therefore balance transparency, robustness, and the ability to capture subtle, nonlinear relationships.

Features and Preprocessing

The feature set is drawn exclusively from company-reported fundamentals. This includes Sales, Expenses, Operating Profit, OPM%, Other Income, Interest, Depreciation, Profit Before Tax, Tax %, Net Profit, and EPS. For BFSI firms, sector-specific disclosures—Revenue, Financing Profit, and Financing Margin %—are retained without modification. Two categorical context variables, Sector (IT, BFSI, FMCG, Pharma, Auto) and Capitalisation category (Large/Mid/Small), are incorporated to reflect heterogeneity across industries and firm sizes.

Deliberate exclusions are made to prevent leakage: identifiers (Company, Ticker), all calendar dates, and any variables derived from post-announcement prices (Closing Price T, T+1 to T+3, 3-Day Return, Movement Label). The models are therefore trained only on information that would have been available to investors at the time of the earnings announcement.

Preprocessing follows a sector-sensitive logic. Missing values are imputed using sector medians for numerical features and modes for categorical features, ensuring imputation respects industry-specific distributions. Numerical features are standardized with **StandardScaler** to equalize scales across variables of very different magnitudes. Categorical variables are encoded with one-hot encoding, using a drop-first approach to avoid redundancy. All transformations are implemented within **scikit-learn Pipelines**, which guarantees that preprocessing occurs inside each cross-validation fold and prevents leakage of information across train–test boundaries.

Choice of Models

Three classifiers are employed, representing a progression from simple and interpretable to complex and powerful:

- **Logistic Regression** serves as the linear baseline, interpretable; `class_weight="balanced"`, L2 regularization tuned. It assumes additive log-odds effects of financial indicators on the Up/Down outcome. While simple, it provides a clear benchmark and interpretable coefficients that can be directly linked to fundamentals.
- **Random Forest** is chosen for its ability to handle collinearity and capture nonlinear interactions. By averaging across many decorrelated decision trees, Random Forests can detect threshold effects and sector-dependent conditions that linear models cannot. Their embedded feature importance metrics, such as permutation importance, directly support RQ2 on identifying influential indicators.
- **XGBoost** represents a state-of-the-art boosting ensemble. It builds trees sequentially, correcting errors iteratively, and includes strong regularization to mitigate overfitting in relatively small datasets. Its capacity to capture nuanced, nonlinear dependencies makes it especially appropriate for financial fundamentals, which often exert effects conditionally rather than linearly. XGBoost also integrates with SHAP values, enabling transparent attribution of predictions to individual features—crucial for understanding model decisions in the context of RQ2 and RQ3.

Together, these models form a methodological spectrum: Logistic Regression tests for linear signal, Random Forest generalizes to nonlinear patterns, and XGBoost pursues the strongest possible predictive performance within the structured tabular setting.

Evaluation Framework

All models are evaluated within a consistent framework. **Stratified 5-fold cross-validation** ensures that each fold maintains the overall Up/Down distribution ($\approx 41\%/59\%$). The **primary metric is ROC AUC**, as it directly measures discrimination between positive and negative cases while being robust to class imbalance. Supporting metrics—accuracy, precision, recall, and F1-score—are also reported for interpretability in business contexts. To address RQ3, the evaluation is stratified by sector: performance metrics are computed separately for IT, BFSI, FMCG, Pharma, and Auto. While sector-specific sample sizes are smaller, this approach highlights whether predictive strength is concentrated in certain industries. In parallel, feature importance outputs (via permutation for Random Forest and SHAP for XGBoost) provide evidence on which fundamentals matter most, both globally and within individual sectors.

Preliminary Results and Interim Reflections

Preliminary Results and Interim Reflections

The interim results provide the first systematic evidence on whether structured fundamentals alone can predict post-earnings price direction in Indian equities. While predictive power remains modest, the findings illuminate important dynamics that will guide the final phase of this research.

Aggregate performance (RQ1, RQ4)

Using 5-fold shuffled cross-validation on pooled data, all three candidate models delivered only small lifts over chance. Logistic Regression generalized best, achieving

Accuracy ≈ 0.57 , F1 ≈ 0.26 , and ROC-AUC ≈ 0.55 with low overfitting. Random Forest and XGBoost, by contrast, fit the training data almost perfectly but collapsed out-of-sample, with test ROC-AUCs of ≈ 0.48 – 0.49 . These results confirm RQ1: structured fundamentals do contain weak but non-zero signal at a three-day horizon. For RQ4, they also show that the linear baseline is more reliable than complex ensembles when using raw financial disclosures.

📊 Key Metrics: Accuracy, F1-Score, ROC-AUC, Overfitting Gap

Model	Metric	Baseline	Tuned	Change
Logistic_Regression	Accuracy	0.572	0.586	+0.014
	F1-Score	0.259	0.008	-0.251
	ROC-AUC	0.552	0.547	-0.005
	Overfitting	0.035	0.007	+0.028
Random_Forest	Accuracy	0.495	0.581	+0.086
	F1-Score	0.334	0.132	-0.202
	ROC-AUC	0.484	0.527	+0.044
	Overfitting	0.505	0.062	+0.444
XGBoost	Accuracy	0.508	0.570	+0.062
	F1-Score	0.373	0.127	-0.246
	ROC-AUC	0.486	0.527	+0.041
	Overfitting	0.492	0.082	+0.410

Figure X : Accuracy, F1-Score, ROC-AUC, Overfitting Gap

Figures X illustrate this pattern: Logistic Regression consistently outperforms the chance line, while RF/XGB add variance without delivering higher mean AUC.

Sector-wise performance (RQ3)

When models are tuned and evaluated by sector, heterogeneity becomes clear. IT/Technology and Auto stand out: Logistic Regression reached F1 ≈ 0.64 in IT, while XGBoost achieved ≈ 0.50 in Auto. In contrast, BFSI proved most challenging, with all models struggling—Logistic Regression collapsed to predicting the majority class (F1

≈ 0.00), while RF and XGB reached only ≈ 0.31 – 0.23 . FMCG and Pharma produced intermediate results, with no model clearly dominant. This validates RQ3: predictive strength is not uniform, but sector conditioned.

SECTOR PERFORMANCE RANKING:		

1. IT_Technology	: Mean F1=0.613, Max F1=0.647	(3 models)
2. Auto	: Mean F1=0.452, Max F1=0.500	(3 models)
3. Pharma	: Mean F1=0.425, Max F1=0.462	(3 models)
4. FMCG	: Mean F1=0.410, Max F1=0.483	(3 models)
5. BFSI	: Mean F1=0.184, Max F1=0.316	(3 models)
MODEL PERFORMANCE RANKING:		

1. XGBoost	: Mean F1=0.436, Max F1=0.629	(5 sectors)
2. LogisticRegression	: Mean F1=0.407, Max F1=0.647	(5 sectors)
3. RandomForest	: Mean F1=0.407, Max F1=0.562	(5 sectors)

Figure X : Sector and Model performance Ranking

Drivers of predictions (RQ2)

Explainability analyses using permutation importance and SHAP reinforce the sectoral story. Across models, profitability and financing structure variables (OPM%, Net Profit, Other Income, Interest, PBT, Revenue) consistently emerge as the most influential. However, their effects are weak and context dependent. For example, high EPS values do not guarantee “Up” predictions; in some sectors they correlate with “Down,” reflecting disappointment relative to expectations. This answers RQ2: fundamentals do matter, but their predictive weight is conditional on sector and context rather than absolute levels.

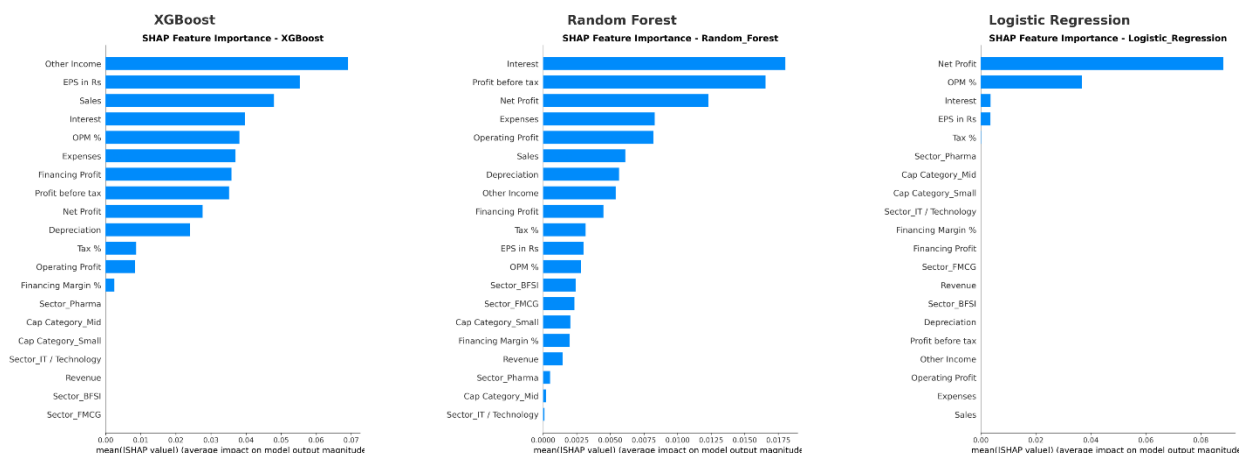


Figure X : SHAP comparison across Models.

Interim conclusion

The preliminary results confirm that structured financial disclosures alone offer only limited predictive power. However, the consistency of weak but non-random signals, combined with strong sectoral variation and identifiable feature importance, justifies further modelling. The next stage will introduce engineered features such as quarter-on-quarter deltas, surprise measures, profitability ratios, sector scaling, stronger regularization, and sector-aware evaluation. These refinements are expected to sharpen the faint signals observed here into more actionable models.

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Appendix

A1. Data Sources

- **Fundamentals and Result Dates:** IndianAPI quarterly results and board meetings endpoints.
- **Daily Prices:** BSE/NSE Bhav Copy archives; IndianAPI daily endpoint (validated); Alpha Vantage TIME_SERIES_DAILY (fallback).
- **GitHub Repository:** [abhivedwa/Walsh-MS_CapstoneProject](https://github.com/abhivedwa/Walsh-MS_CapstoneProject) contains data cleaning, EDA, and modelling notebooks.

A2. Dataset Overview

- **Sample:** 584 firm-quarters (July 2022–July 2025) across five sectors.
- **Target:** *Movement Label* (Up = 1, Down = 0), defined by 3-day post-earnings return.
- **Key Features:** Sales, Net Profit, EPS, OPM%, Interest, Depreciation, PBT, BFSI-specific margins.

- **Derived Metrics:** 3-Day Avg Price, 3-Day Return (%).

A3. Modelling Framework

- **Algorithms:** Logistic Regression, Random Forest, XGBoost.
- **Validation:** 5-fold stratified cross-validation.
- **Metrics:** ROC AUC (primary), Accuracy, Precision, Recall, F1-score.
- **Explainability:** Random Forest permutation importance; XGBoost SHAP values.

A4. Supplementary Results (Summary)

- Logistic Regression yielded modest but stable results (ROC AUC ≈ 0.55).
- Random Forest and XGBoost overfit in pooled analysis but showed sector-specific strengths (e.g., IT, Auto).
- Profitability and financing ratios (EPS, OPM%, Net Profit, Interest) emerged as the most influential predictors.

Table A1

Summary of pooled model performance (cross-validated)

Model	Accuracy	F1	ROC AUC
Logistic Regression	0.57	0.26	0.55
Random Forest	0.50	0.33	0.48
XGBoost	0.51	0.37	0.49

A5. Glossary of Financial Terms

- **Short-Term Reaction:** Price movements observed within days after earnings announcements (here: 3 trading days).

- **Long-Term Trend:** Price behaviour across multiple quarters or years, often driven by fundamentals and broader sentiment.
 - **EPS (Earnings per Share):** Net Profit divided by total shares outstanding; a profitability measure.
 - **OPM% (Operating Profit Margin):** Operating Profit divided by Sales; captures operational efficiency.
 - **PAT (Profit After Tax):** Net income after tax; also called Net Profit.
 - **PBT (Profit Before Tax):** Profit before accounting for income taxes.
 - **Revenue:** Top-line income from operations; in BFSI, refers to interest/financing income.
 - **Financing Margin % (NIM):** Net Interest Margin; BFSI measure of lending profitability.
 - **Short-Term Return (%):** Percentage change in stock price over 3 days post earnings.
 - **Movement Label:** Binary indicator (1 = Up $\geq 1\%$ return, 0 = Down $< 1\%$).
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A6. Reproducibility Note

All preprocessing, analysis, and modelling steps are documented in the GitHub repository:

 [abhivedwa/Walsh-MS_CapstoneProject](https://github.com/abhivedwa/Walsh-MS_CapstoneProject)